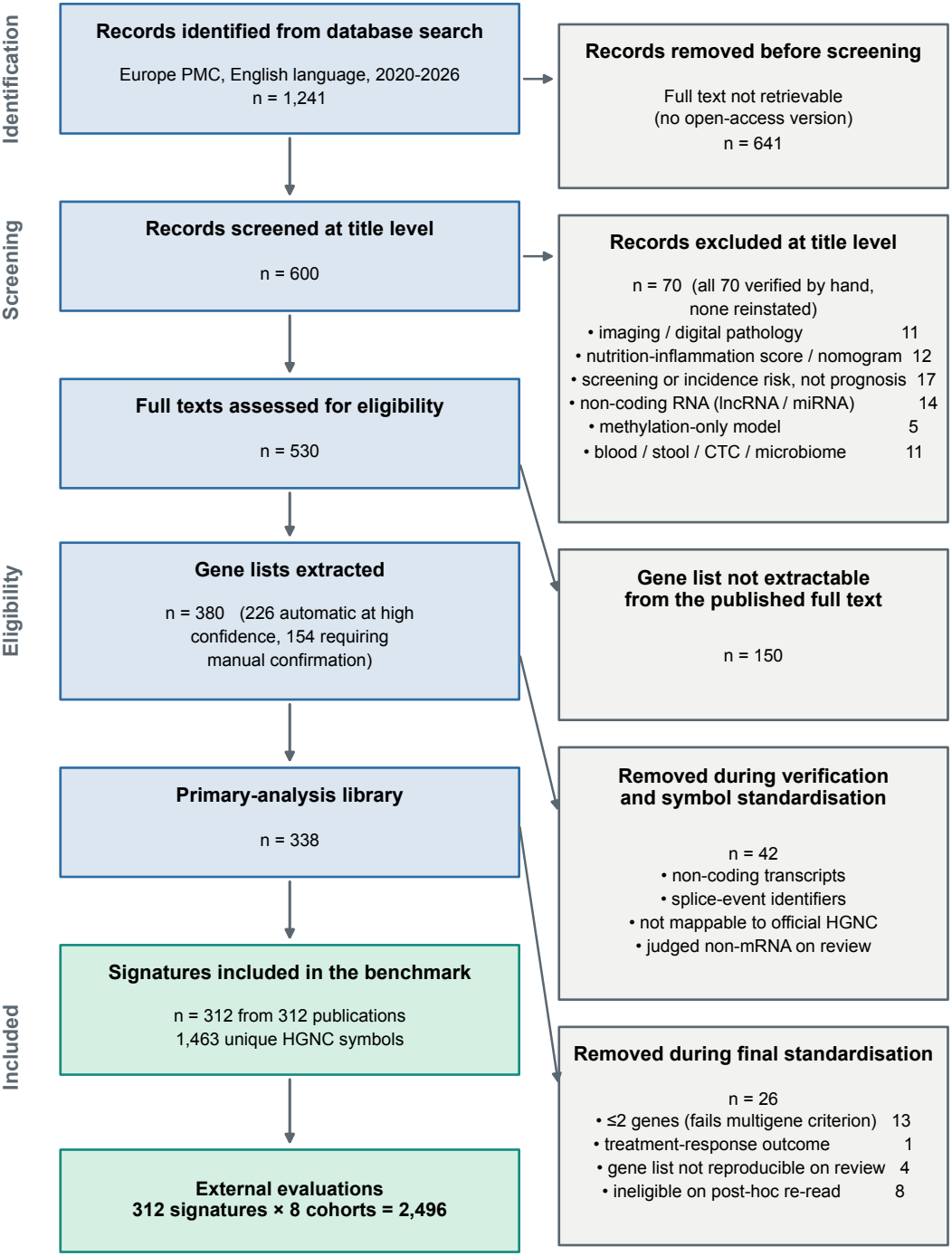
Identification
Records identified from database search
Europe PMC, English language, 2020-2026
n = 1,241
Records removed before screening
Full text not retrievable
(no open-access version)
n = 641
Screening
Records screened at title level
n = 600
Records excluded at title level
n = 70 (all 70 verified by hand, none reinstated)
• imaging / digital pathology 11
• nutrition-inflammation score / nomogram 12
• screening or incidence risk, not prognosis 17
• non-coding RNA (lncRNA / miRNA) 14
• methylation-only model 5
• blood / stool / CTC / microbiome 11
Full texts assessed for eligibility
n = 530
Eligibility
Gene lists extracted
n = 380 (226 automatic at high confidence, 154 requiring manual confirmation)
Gene list not extractable from the published full text
n = 150
Primary-analysis library
n = 338
Removed during verification and symbol standardisation
n = 42
• non-coding transcripts
• splice-event identifiers
• not mappable to official HGNC
• judged non-mRNA on review
Included
Signatures included in the benchmark
n = 312 from 312 publications
1,463 unique HGNC symbols
Removed during final standardisation
n = 26
• ≤2 genes (fails multigene criterion) 13
• treatment-response outcome 1
• gene list not reproducible on review 4
• ineligible on post-hoc re-read 8
External evaluations
312 signatures × 8 cohorts = 2,496